UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 11 Physics Final Exam Practice

Term 1 2026–2027 • Set B • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. Convert 90° to radians. [4 marks]

A. 2π rad
B. π rad
C. π/4 rad
D. π/2 rad

Reference: pp. 3–4 | **Answer D**

2. Two points on one rigid rotating disk have radii r and 2r. Their tangential speeds have ratio: [4 marks]

A. 1 : 1
B. 1 : 2
C. 2 : 1
D. 1 : 4

Reference: pp. 4–5 | **Answer B**

3. A trailer wheel has diameter 0.48 m and linear acceleration 1.85 m/s². Its angular acceleration is: [4 marks]

A. 7.71 rad/s²
B. 0.444 rad/s²
C. 3.85 rad/s²
D. 0.130 rad/s²

Reference: p. 6, Practice Problems 3–4 | **Answer A**

4. A washer rotating at 635 rev/min stops uniformly in 8.0 s. Its angular acceleration is approximately: [4 marks]

A. +8.31 rad/s²
B. −0.132 rad/s²
C. −8.31 rad/s²
D. −79.4 rad/s²

Reference: p. 6, Check Your Progress 8 | **Answer C**

5. Which statement is true for points at different nonzero radii on a rigid rotating disk? [4 marks]

A. Their circular paths have equal lengths
B. The outer point takes longer to complete one revolution
C. Their angular speeds are equal
D. Their tangential speeds are always equal

Reference: pp. 4–5 | **Answer C**

6. A torque of 55.0 N m is needed and the force is 135 N. The required perpendicular lever arm is: [4 marks]

A. 7425 m
B. 0.407 m
C. 0.245 m
D. 2.45 m

Reference: p. 9, Practice Problem 12 | **Answer B**

7. For a two-mass baton (0.30 kg at each end, length 0.66 m), I about one end is: [4 marks]

A. 0.396 kg m²
B. 0.0131 kg m²
C. 0.131 kg m²
D. 0.0653 kg m²

Reference: p. 14, Example Problem 3 | **Answer C**

8. A wide base improves stability because: [2 marks]

A. Gravity acts horizontally
B. The mass becomes zero
C. All torques disappear in every orientation
D. The centre of mass can shift farther before its vertical line leaves the base

Reference: p. 24, Check Your Progress 42, 44–45 | **Answer D**

9. A body has zero net force but nonzero net torque. It: [4 marks]

A. Must be in static equilibrium
B. Cannot rotate
C. Can have angular acceleration
D. Must have zero angular acceleration

Reference: pp. 20–21 | **Answer C**

10. Viewed from Earth, a long-range moving object can appear to curve because Earth: [4 marks]

A. Stops rotating during the motion
B. Exerts no gravity
C. Changes the object's mass
D. Rotates while the object moves

Reference: pp. 23–24 | **Answer D**

11. Two balls collide over the same contact interval. Their interaction impulses are: [4 marks]

A. Unrelated to their momentum changes
B. Equal in magnitude and opposite in direction
C. Zero for the heavier ball
D. Always equal in direction

Reference: pp. 30–31 | **Answer B**

12. A 7.0 kg object initially moves at 2.0 m/s. A +5.0 N force acts for 1.0 s. Its new speed is: [4 marks]

A. 7.00 m/s
B. 0.714 m/s
C. 1.29 m/s
D. 2.71 m/s

Reference: p. 33, Practice Problem 3 | **Answer D**

13. A 1.5 kg cart moves left at 0.27 m/s after a spring pushes it away from a 4.5 kg cart. Initially both were at rest. The second cart moves: [4 marks]

A. Right at 0.090 m/s
B. Left at 0.81 m/s
C. Right at 0.81 m/s
D. Left at 0.090 m/s

Reference: p. 43, Practice Problem 22 | **Answer A**

14. For the same impulse, doubling the collision time makes the average force: [4 marks]

A. Half as large
B. Unchanged
C. Twice as large
D. Four times as large

Reference: pp. 30–33 | **Answer A**

15. A diver tucks the body during a somersault. The rotation speeds up because: [4 marks]

A. The moment of inertia decreases at nearly constant angular momentum
B. The angular momentum vanishes
C. The mass increases
D. A large external torque must appear

Reference: pp. 34–35 | **Answer A**

16. Two players approach from opposite directions, collide and come to rest. Neglect horizontal external impulse. Initially their momenta were: [2 marks]

A. Unrelated
B. Equal in magnitude and opposite in direction
C. Both zero
D. Equal in direction

Reference: p. 47, Check Your Progress 32 | **Answer B**

Set B Section B Free Response

17. A film lasts 2.0 h. Consider the hands of an analogue clock during this interval. Separately, a 43 kg child sits 1.8 m from the pivot of a light seesaw; a 52 kg child sits on the other side. Take counterclockwise as positive. [10 marks]

Reference: p. 6, Check Your Progress 7, 9; p. 12, Practice Problem 16; pp. 20–21

(a) Find the signed angular displacement of the hour and minute hands as viewed from the front. [3]

Answer Both move clockwise, hence negative [1]; hour hand $-\pi/3$ rad [1]; minute hand -4π rad [1].

(b) Do all points on the minute hand have equal angular displacement and travel equal distances? Explain. [2]

Answer Equal angular displacement [1]; different distances because $s = r\theta$ [1].

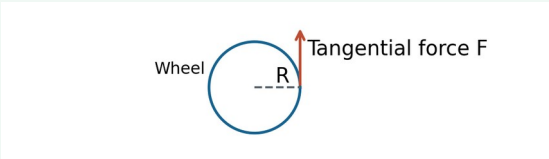
(c) Calculate the distance from the pivot at which the 52 kg child must sit for balance. [3]

Answer $43g(1.8) = 52g\ r$ [1]; $r = 43(1.8)/52$ [1] = 1.49 m [1].

(d) Why must both net force and net torque vanish for static equilibrium? [2]

Answer Zero net force prevents translational acceleration [1]; zero net torque prevents angular acceleration [1].

18. A solid uniform wheel has mass 5.2 kg and diameter 0.55 m. Starting from rest, it must reach 12 rev/s in 35 s. A strap pulls tangentially at its rim. Use $I = \frac{1}{2}MR^2$. [10 marks]



Reference: p. 17, Check Your Progress 33; pp. 34–36

(a) Find its moment of inertia. [2]

Answer R = 0.275 m [1]; $I = \frac{1}{2}(5.2)(0.275)^2 = 0.1966$ kg m² [1].

(b) Find angular acceleration and required torque. [3]

Answer $\omega_f = 24\pi = 75.4$ rad/s [1]; $\alpha = 75.4/35 = 2.15$ rad/s² [1]; $\tau = I\alpha = 0.424$ N m [1].

(c) Find the strap force. [2]

Answer $F = \tau/R$ [1] = 0.424/0.275 = 1.54 N [1].

(d) Find final angular momentum and the angular impulse. [3]

Answer $L_f = I\omega_f$ [1] = 14.8 kg m²/s [1]; angular impulse = ΔL = +14.8 N m s [1].

19. A 240 kg snowmobile speeds up from 6.0 m/s to 28.0 m/s in the same direction over 60.0 s. Take that direction as positive. [10 marks]

Reference: p. 33, Practice Problem 4; pp. 30–31

(a) Calculate its initial and final momenta. [3]

Answer $p = mv$ [1]; p_i = +1440 kg m/s [1]; p_f = +6720 kg m/s [1].

(b) Calculate momentum change and impulse. [2]

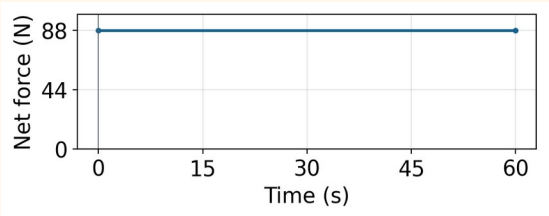
Answer $\Delta p = 6720 - 1440 = +5280$ kg m/s [1]; J = +5280 N s [1].

(c) Find the average net force and represent it on a constant force–time graph. [3]

Answer $F_{average} = \Delta p/\Delta t = 88$ N [1]; labelled t (s) and F (N) axes [1]; horizontal 88 N line over 0–60 s [1].

(d) Explain why the area under this graph equals the momentum change. [2]

Answer Area = $F\Delta t = 88(60) = 5280$ N s [1]; impulse–momentum theorem gives $J = \Delta p$ [1].



20. An astronaut and thruster pistol have combined mass 84 kg after a burst of gas is released. Initially the system is at rest. The burst has mass 35 g and velocity −875 m/s. Neglect external impulse. [10 marks]

Reference: p. 42, Example Problem 4; pp. 38–42

(a) Define a suitable system and state its initial momentum. [2]

Answer Astronaut, pistol and released gas [1]; $p_i = 0$ [1].

(b) Find the gas momentum and the astronaut's recoil momentum. [3]

Answer m_{gas} = 0.035 kg [1]; p_{gas} = −30.625 kg m/s [1]; $p_{astronaut}$ = +30.625 kg m/s [1].

(c) Calculate the astronaut's velocity and direction. [3]

Answer $0 = 84v + 0.035(-875)$ [1]; v = +0.365 m/s [1], opposite the gas [1].

(d) Explain why the thruster works in space without air and state the astronaut's impulse. [2]

Answer Internal forces give opposite momentum changes; no surrounding air is required [1]. J = +30.625 N s [1].